



RoToR: Towards More Reliable Responses for Order-Invariant Inputs

Soyoung Yoon^{1*} **Dongha Ahn¹²** **Youngwon Lee¹** **Minkyu Jung²**
HyungJoo Jang² **Seung-won Hwang^{1†}**

¹Seoul National University ²Channel Corporation
{soyoung.yoon, seungwonh}@snu.ac.kr



Channel Talk

Overview

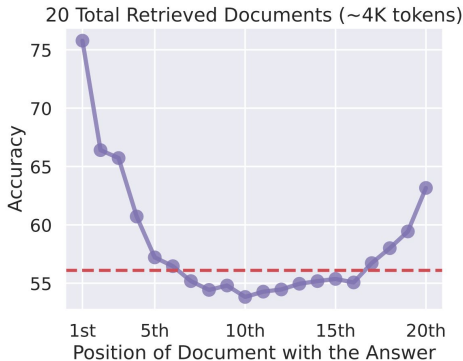
We introduce **RoToR**: a zero-shot order-invariant Language Model, with (1) Global Sorting + Circular Position IDs and (2) Selective Routing for Mixed Inputs, which achieve SOTA robustness on 3 benchmarks and lower FLOPs v.s. Baselines (PINE)

1. Motivation: positional bias for listwise inputs
2. Limitations of prior works
3. Contributions of **RoToR** with **Selective Routing**
4. Experimental results

Motivation: Positional Bias for listwise inputs

- Lost-in-the-Middle (RAG)
- First-choice bias (75%) in LLM-as-a-judge
- MMLU rank shifts by 8 with shuffle
- Need neutral handling for sets, tables, multiple-choice questions

Which one is red?	Which one is red?
A. Apple	A. Orange
B. Orange	B. Apple
C. Grape	C. Grape
Answer: ☒	Answer: ☒
-> A. Apple	-> A. Orange



Liu et al., Lost in the Middle: How Language Models Use Long Contexts

Zheng et al., Judging llm-as-a-judge with mt-bench and chatbot arena.

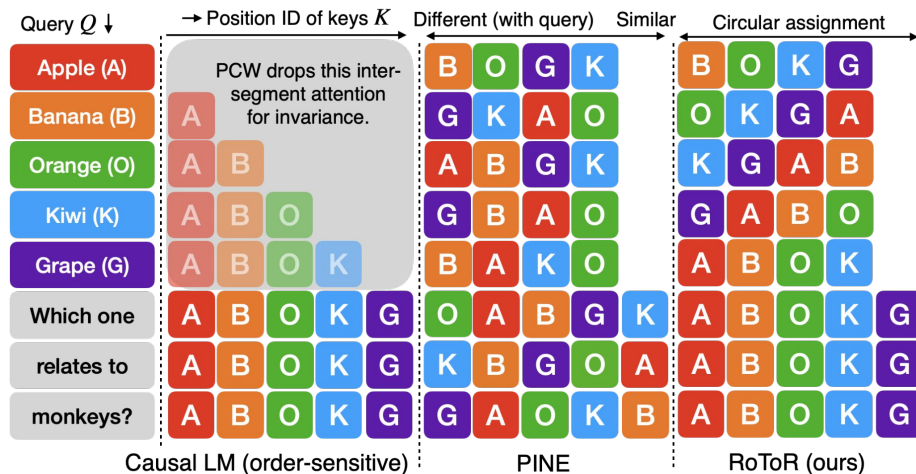
Alzahrani et al., When Benchmarks are Targets: Revealing the Sensitivity of Large Language Model Leaderboards

Limitations of prior works

-> Zero-shot order-invariant LMs have been proposed, but had limitations in two aspects:

1. Training and inference distribution mismatch

- PCW, Set-based prompting: No cross-segment context
- PINE: per-query sort -> $O(O(n^2) + \text{instability})$
- Frequent ID changes cause **OOD behavior** -> drops its ability



<- Self-attention patterns
(x = query, y = key)
across order-invariant
models

Limitations of prior works

-> Zero-shot order-invariant LMs have been proposed, but had limitations in two aspects:

2. Fail to extend to real-life scenarios (order-invariant + order-sensitive)

- Does not consider hybrid cases (e.g., MMLU)
- Cannot mix order-sensitive segments

In 8085 name/names of the 16 bit registers is/are:

Order sensitive

- A. stack pointer
- B. program counter
- C. both A and B.
- D. none of these

Order invariant

- A. stack pointer
- B. program counter
- C. accumulator
- D. microprocessor

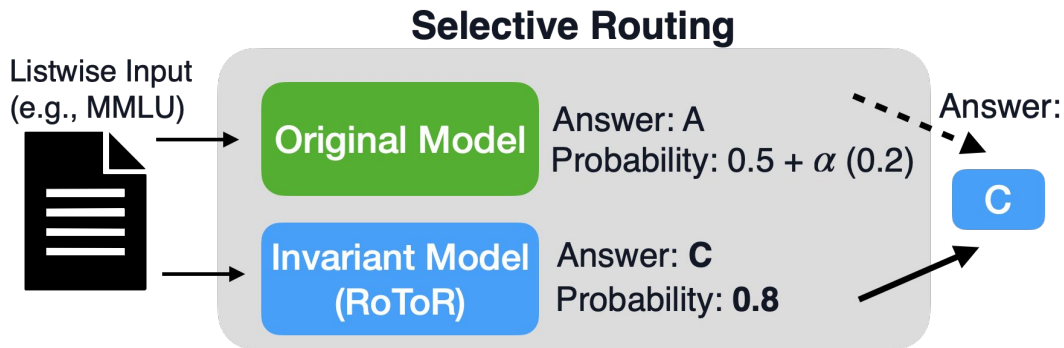
RoToR - Key Contributions

1. Training and inference distribution mismatch

- Stable, order-invariant solution (RoToR)
- Query-agnostic global ordering with minimal positional ID modifications

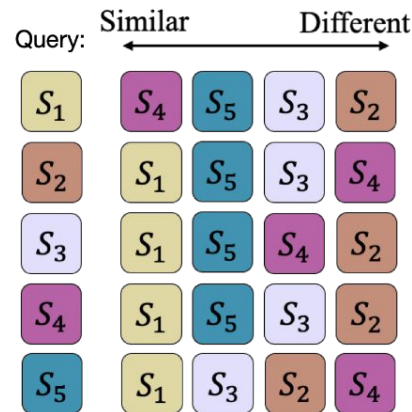
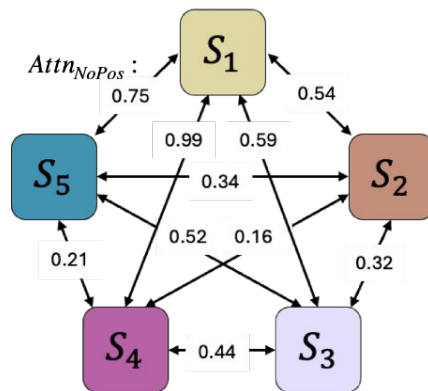
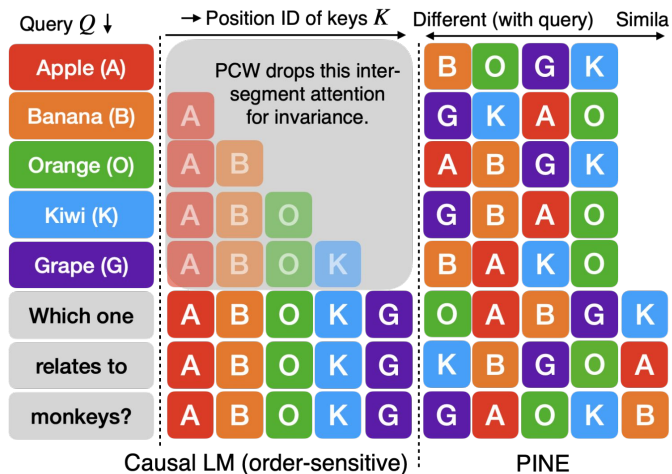
2. Fail to extend to hybrid cases

- Selective Routing, which switches between original / invariant LMs based on confidence



Methodology: Order-invariant causal LMs

- PINE: Bidirectional processing with Q-K similarity
 - Has to obtain the same attention representation, regardless of initial ordering of segments
 - Places query IDs last, sorts other segments in a order-invariant way



Methodology: Order-invariant causal LMs

- PINE: Bidirectional processing with Q-K similarity
 - Has to obtain the same attention representation, regardless of initial ordering of segments
 - Places query IDs last, sorts other segments in a order-invariant way

Sorting Criterion: KV attention value *without* position ID



Attention matrix of original model

Attention matrix of PINE

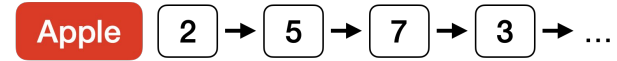
PINE, shown by position id

Methodology: Order-invariant causal LMs

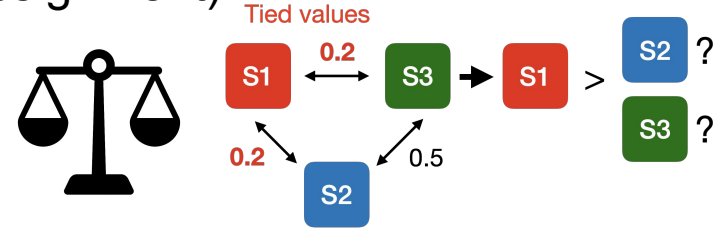
- PINE: Challenges

- Frequent alterations on position IDs (layer, head, suffix & generation tokens)
- Computationally expensive (per-query KV attention compute)
- Numerical Instability (arising from attention assignment)

Position ID:



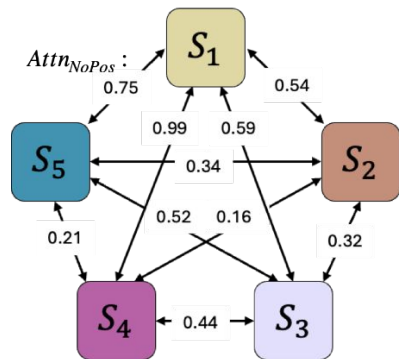
- Solution: RoToR



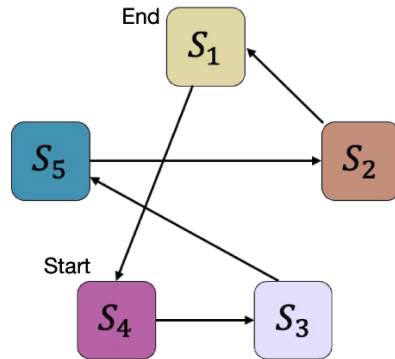
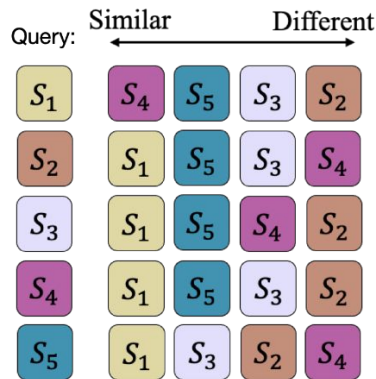
- Keep the bidirectional structure, but alter the position assignment in a simple and stable way!
- Define a single global ordering + circular arrangement

RoToR v.s. PINE (Schematic)

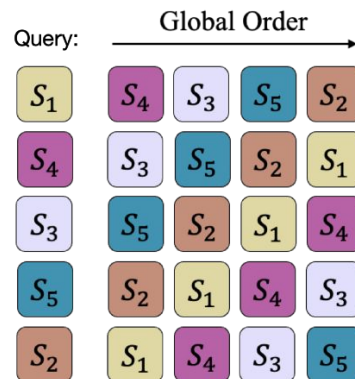
PINE: query-dependent grid, RoToR: fixed order, rotate per query
-> Stable IDs, zero collisions, less computation



PINE

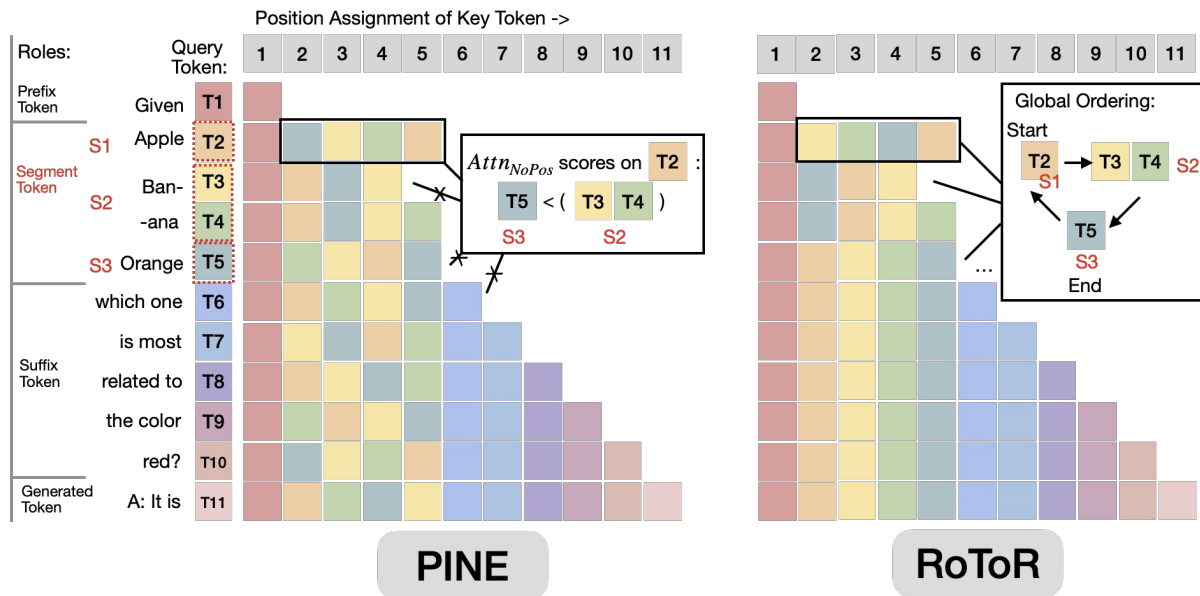


RoToR



RoToR v.s. PINE (Schematic)

PINE: query-dependent grid, RoToR: fixed order, rotate per query
-> Stable IDs, zero collisions, less computation



RoToR v.s. PINE (Schematic)

We propose three different global ordering methodology:

1. lexical (base)

- Hierarchical sort on tokenized ID numbers

2. Reranking (monot5)

3. Frequency-normalized

Different global ordering strategies given example input: 7 rows

0: row content : direct war losses | 360,000 | null | null | null | null | 183,000 | 543,000!!!!!!!!!!!!!!
 1: row content : murdered | 75,000 | 100,000 | 116,000 | 133,000 | 82,000 | null | 506,000!!!!!!
 2: row content : deaths in prisons & camps | 69,000 | 210,000 | 220,000 | 266,000 | ...
 3: row content : deaths outside of prisons & camps | null | 42,000 | 71,000 | 142,000 | ...
 4: row content : murdered in eastern regions | null | null | null | null | null | 100,000 | 100,000!!!!!!!!!!!!!!
 5: row content : deaths other countries | null | null | null | null | null | null | 2,000!!!!!!!!!!!!!!
 6: row content : total | 504,000 | 352,000 | 407,000 | 541,000 | 681,000 | 270,000 | 2,770,000!

↓ Tokenized IDs

(1) Global sort as **lexical sort**: 0->6->2->5->3->4->1

0 [198, 654, 2262, 551, 2167, 4288, 18151, 765, 220, 6843, 11 ...]
 1 [198, 654, 2262, 551, 32485, 765, 220, 2075, 11, 931, 765 ...]
 2 [198, 654, 2262, 551, 16779, 304, 49429, 612, 29135, 765, 2 ...]
 3 [198, 654, 2262, 551, 16779, 4994, 315, 49429, 612, 29135, 765 ...]
 4 [198, 654, 2262, 551, 32485, 304, 24024, 13918, 765, 854, 7 ...]
 5 [198, 654, 2262, 551, 16779, 1023, 5961, 765, 854, 765, 854 ...]
 6 [198, 654, 2262, 551, 2860, 765, 220, 18048, 11, 931, 765 ...]

Frequency Count:

{765: 10, 198: 7, 654: 7, 2262: 7, 551: 7, 220: 3, 11: 3, 16779: 3, 854: 3, 32485: 2, 931:...}

Divide by
Frequency

(3) Global sort as **frequency-based sort**: 2->5->3->1->4->6->1

0 [1/7, 1/7, 1/7, 1/7, 1/1, 1/1, 1/1, 1/10, 1/3, 1/1, 1/3, ...]
 1 [1/7, 1/7, 1/7, 1/7, 1/2, 1/10, 1/3, 1/1, 1/3, 1/2, 1/10, ...]
 2 [1/7, 1/7, 1/7, 1/7, 1/3, 1/2, 1/2, 1/2, 1/2, 1/10, 1/1, ...]
 3 [1/7, 1/7, 1/7, 1/7, 1/3, 1/1, 1/1, 1/2, 1/2, 1/2, 1/10, ...]
 4 [1/7, 1/7, 1/7, 1/7, 1/2, 1/2, 1/1, 1/1, 1/10, 1/3, 1/1, ...]
 5 [1/7, 1/7, 1/7, 1/7, 1/3, 1/1, 1/1, 1/10, 1/3, 1/10, 1/3, ...]
 6 [1/7, 1/7, 1/7, 1/7, 1/1, 1/10, 1/3, 1/1, 1/3, 1/2, 1/10, ...]

(2) Global sort as **reranking sort**:

0: 0.1
 1: 0.15
 2: 0.4
 3: 0.05
 4: 0.2
 5: 0.6
 6: 0.9

← $\frac{p(\text{Yes})}{p(\text{Yes}) + p(\text{No})}$

6->5->2->4->1->0->3

Query: what is the total number of murdered?, Document: row content : direct war losses | 360... 3,000!!!!!!!!!!!!!!!, Relevant:

...

Query: what is the total number of murdered?, Document: row content : total | 504,000 | 35...000 | 2,770,000!, Relevant:

Reranker
(MonoT5-base)

Figure 6: Illustration of ordering 7 rows by 2 different global sort options, (1) lexical sort based on token ids, or (2) reranking sort based on a reranker model (MonoT5-base in this case). (3) frequency based sorting applies lexical sorting, but the definitive token ids are mapped into the inverse frequency.

Selective Routing: extend to hybrid cases (e.g., MMLU)

- Compute confidence of Original & RoToR outputs
- Choose higher $p + \alpha$ ($\alpha=0.2$)

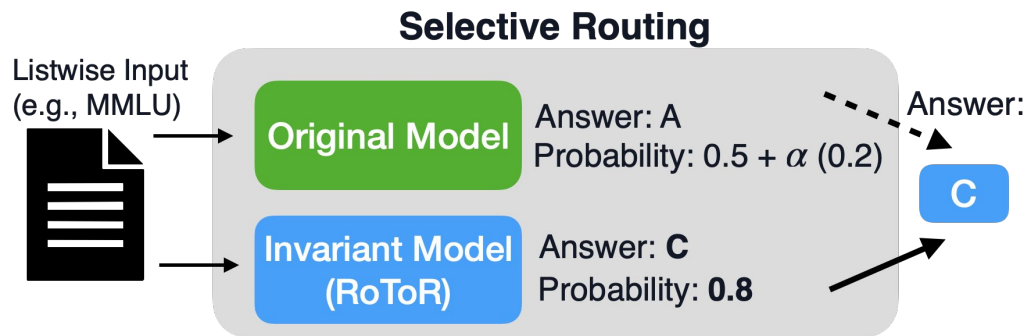
In 8085 name/names of the 16 bit registers is/are:

Order sensitive

- A. stack pointer
- B. program counter
- C. both A and B.
- D. none of these

Order invariant

- A. stack pointer
- B. program counter
- C. accumulator
- D. microprocessor



Experimental Setup

- **Benchmarks:**

- Lost-in-the-Middle (LitM)
- Knowledge Graph QA (KGQA): Mintaka
- MMLU: selective routing cases
- LongBench: long context scenarios (Appendix)

- **Model backbones:**

- Llama-3.1-8B/70B
- Qwen-1.5-4/7/72B-Chat

- **Metrics:** best_subspan_em (LitM), EM, F1, Acc. (KGQA), Acc. (MMLU)

- **Methods:** Original (order-sensitive), PCW, Set-based prompting, PINE, RoToR

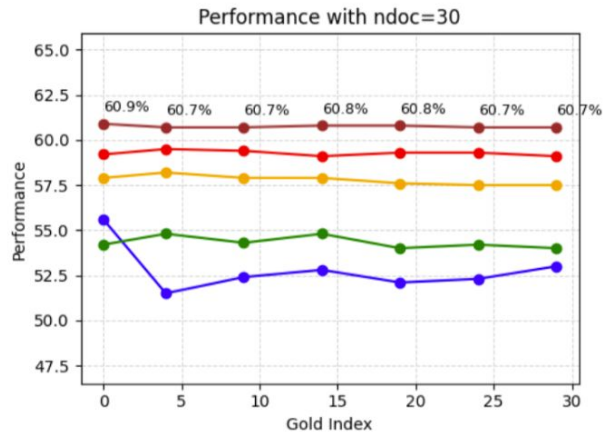
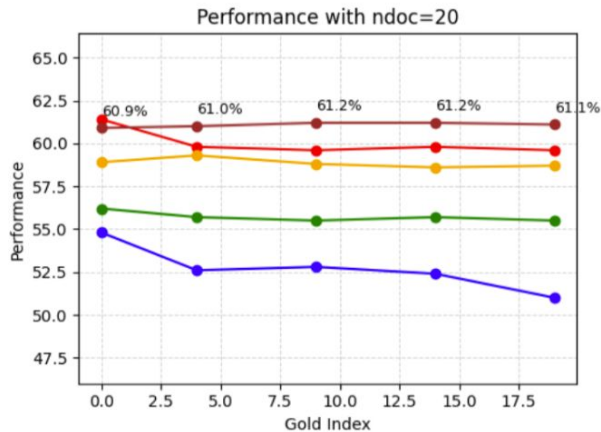
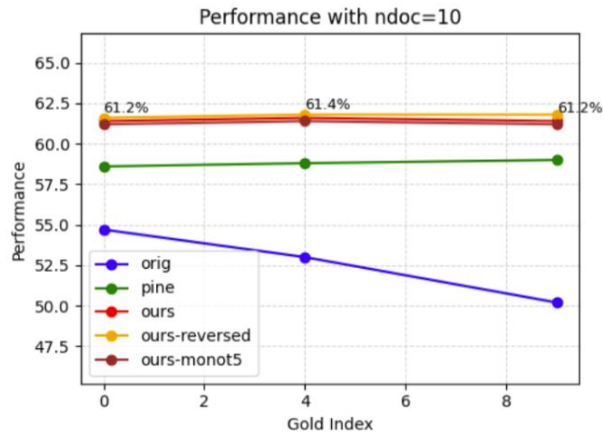
Efficiency Gains (over PINE)

- Less computation:
 - Overhead FLOPs $\downarrow$ 98 % (72B)
- Faster:
 - E2E Latency $\downarrow$ 23-43 % on LitM
- Reduces OOD:
 - Perplexity $\downarrow$; collision rate 0 %

Model	Benchmark	PINE	RoToR	Reduction
(a) Overhead FLOPs, relative to original model				
Llama-3.1-8B-Instruct	MMLU, $N = 4$	$0.59\times$	$0.55\times$	7.6%
	LitM, $N = 10$	$7.07\times$	$4.81\times$	31.9%
	LitM, $N = 30$	$22.43\times$	$15.05\times$	32.9%
Llama-3.1-70B-Instruct	KGQA, $N = 30$	$1.27\times$	$0.94\times$	26.0%
	KGQA, $N = 50$	$1.82\times$	$1.29\times$	29.0%
Qwen1.5-72B-Chat	KGQA, $N = 30$	$0.45\times$	$0.01\times$	98.0%
	KGQA, $N = 50$	$0.58\times$	$0.03\times$	94.8%
(b) End-to-end latency (s)				
Llama-3.1-70B-Instruct	LitM, $N = 10$	57,352	44,219	22.9%
	LitM, $N = 20$	87,091	58,680	32.6%
Llama-3.1-8B-Instruct	MMLU, $N = 4$	7,371	6,608	10.4%
	LitM, $N = 10$	18,551	14,264	23.1%
	LitM, $N = 30$	41,664	23,569	43.4%
(c) Perplexity & Collision rate, (on LitM)				
Llama-3.1-8B-Instruct	Perplexity ($N = 20$)	6.91	6.65	–
	Collision rate ($N = 30$)	42.3%	0 (None)	–

Table 4: **Unified efficiency comparison of RoToR vs. PINE**, reporting (a) Additional FLOPs, (b) Latency, and (c) Perplexity & Collision rate. Columns list each metric for PINE and RoToR, and the relative reduction. Yellow rows separate sub-sections.

Results: Lost-in-the Middle (LitM)



- Original Model fluctuates performance
- Ours (RoToR): maintains stable & higher performance than other order-invariant models

Results:
Lost-in-the
Middle
(LitM)

- Full results

Total ndoc (segments)	10			20					30						
Gold idx at:	0	4	9	0	4	9	14	19	0	4	9	14	19	24	29
Llama-3.1-8B-Instruct															
Original	54.7	53.0	50.2	54.8	52.6	52.8	52.4	51.0	55.6	51.5	52.4	52.8	52.1	52.3	53.0
PCW	12.4	11.9	12.2	3.7	4.0	4.0	4.0	3.9	2.3	1.8	2.0	2.0	2.1	2.0	2.0
Set-Based Prompting	42.5	42.5	42.5	26.3	26.3	26.3	26.3	26.3	14.1	14.1	14.1	14.1	14.1	14.1	14.1
PINE	58.6	58.8	59.0	56.2	55.7	55.5	55.7	55.5	54.2	54.8	54.3	53.7	54.8	54.2	54.0
RoToR-lexical	61.4	61.6	61.6	61.4	59.8	59.6	59.6	59.8	59.2	59.5	59.4	59.1	59.0	59.3	59.1
RoToR-reversed lexical	61.6	61.8	61.8	58.9	59.3	58.8	58.6	58.7	57.9	58.2	57.9	57.4	57.9	57.6	57.5
RoToR-MonoT5	61.2	61.4	61.2	60.9	61.0	61.2	61.2	61.2	60.9	60.7	60.7	60.7	60.8	60.8	60.7
RoToR-Freq.	61.0	61.1	61.1	60.4	60.3	58.6	60.2	60.0	59.3	60.4	59.7	59.5	59.5	59.6	59.2
Llama 3.1-70B-Instruct															
Original	66.2	65.7	65.7	65.2	64.3	65.0	66.2	64.8	—						
PINE	67.9	67.8	67.5	65.9	65.7	65.9	65.8	65.5	—						
RoToR	69.6	69.5	69.3	67.6	67.8	67.8	67.7	67.9	—						
RoToR-MonoT5	68.9	69.0	68.8	67.5	67.5	67.7	67.5	67.6	—						
Qwen1.5-4B-Chat															
Original	61.3	54.8	53.1	59.5	49.1	47.9	45.9	48.3	56.8	45.6	44.9	44.6	45.3	43.5	48.3
PINE	57.2	57.4	57.0	48.6	48.2	48.2	48.1	48.9	46.4	45.9	46.7	46.6	46.4	46.4	46.3
RoToR	58.5	58.4	58.1	49.9	49.7	49.6	49.8	49.9	44.6	44.8	44.7	44.7	44.9	44.8	44.7
RoToR-MonoT5	58.9	58.5	58.7	52.2	52.1	52.1	52.2	52.6	50.6	50.7	50.5	50.6	50.5	50.6	50.4
RoToR-Freq.	56.7	56.9	56.9	51.9	51.5	51.8	51.6	52.4	46.8	46.7	46.7	46.4	47.0	46.8	46.6
Qwen1.5-7B-Chat															
Original	72.5	63.3	62.9	72.5	58.5	56.1	56.0	58.2	73.1	58.6	55.8	53.3	53.2	52.5	57.5
PINE	65.4	65.5	66.3	59.1	59.4	59.1	58.6	59.2	58.0	55.3	55.7	56.3	55.1	55.8	56.1
RoToR	68.6	68.7	68.6	62.6	62.9	62.7	63.0	62.7	57.0	57.3	59.7	57.4	57.3	62.8	57.0
RoToR-MonoT5	68.8	69.4	69.0	65.2	65.5	65.0	64.9	65.0	62.6	62.8	62.9	62.7	62.9	62.8	62.5
RoToR-Freq.	68.2	68.4	68.4	62.6	62.9	62.8	62.7	62.3	59.5	59.8	59.7	59.6	59.7	59.7	59.7

Results: KGQA

- Top-30 and Top-50 knowledge triples per query
- Test before / after shuffling segments to see robustness
- RoToR obtains lower stdev (better stability) + higher performance than PINE
- Trend persists for > 70B model variants

	Llama-3.1-8B-Instr.			Llama-3.1-70B-Instr.			Qwen1.5-4B-Chat			Qwen1.5-7B-Chat			Qwen1.5-72B-Chat		
Method	Acc.	EM	F1	Acc.	EM	F1	Acc.	EM	F1	Acc.	EM	F1	Acc.	EM	F1
N = 30															
Initial, no shuffling of segments															
Original	50.2	44.0	51.9	61.6	57.7	63.6	30.7	27.9	34.9	31.5	27.8	35.4	41.4	37.7	43.7
PINE	51.5	45.0	52.6	63.1	58.7	64.8	31.6	28.7	35.6	32.3	28.8	36.4	46.7	42.9	49.0
RoToR	53.1	46.5	54.1	63.6	59.1	65.2	32.0	29.0	35.7	34.3	29.8	37.7	47.5	43.2	49.2
RoToR-MonoT5	51.6	45.0	52.5		—		32.3	29.1	36.2	32.9	28.4	36.3		—	
RoToR-Freq.	52.6	46.1	53.7		—		32.3	29.2	36.0	33.7	29.5	37.2		—	
After shuffling segments, averaged over 3 seeds															
Original	49.5	43.3	51.0	62.1	57.8	64.0	30.1	27.5	34.7	31.4	27.3	35.0	41.0	37.6	43.6
↪ stdev. (±)	0.07 / 0.14 / 0.17			0.37 / 0.40 / 0.27			0.41 / 0.34 / 0.43			0.26 / 0.28 / 0.29			0.75 / 0.40 / 0.33		
PINE	51.8	45.2	52.8	63.3	58.8	64.9	31.5	28.7	35.6	32.3	28.8	35.7	46.9	43.3	49.2
↪ stdev. (±)	0.05 / 0.07 / 0.16			0.13 / 0.04 / 0.10			0.20 / 0.18 / 0.13			0.17 / 0.20 / 0.13			0.18 / 0.20 / 0.20		
RoToR	52.8	46.2	53.8	63.5	59.1	65.3	31.8	28.8	35.5	34.2	29.9	37.7	47.4	43.1	49.1
↪ stdev. (±)	0.05 / 0.05 / 0.02			0.11 / 0.07 / 0.08			0.05 / 0.02 / 0.09			0.09 / 0.07 / 0.06			0.06 / 0.04 / 0.07		
RoToR-MonoT5	51.6	45.0	52.6		—		32.4	29.2	36.3	33.0	28.8	36.5		—	
↪ stdev. (±)	0.12 / 0.06 / 0.10			—			0.04 / 0.02 / 0.13			0.12 / 0.09 / 0.07			—		
RoToR-Freq.	52.5	45.9	53.5		—		32.3	29.3	36.0	33.8	29.6	37.4		—	
↪ stdev. (±)	0.10 / 0.15 / 0.11			—			0.13 / 0.16 / 0.09			0.04 / 0.00 / 0.09			—		
N = 50															
Initial, no shuffling of segments															
Original	50.0	44.0	51.7	62.6	58.5	64.5	31.6	28.6	35.8	31.7	28.0	35.7	42.1	38.7	44.5
PINE	51.6	45.1	52.6	64.1	59.8	65.8	31.6	28.8	35.3	32.0	28.5	35.9	48.0	44.1	49.9
RoToR	52.9	46.0	53.6	64.6	60.0	66.2	32.7	29.6	36.2	34.3	30.1	38.0	48.4	44.3	50.3
RoToR-MonoT5	52.4	45.4	52.8		—		32.3	29.3	35.9	32.9	28.9	36.6		—	
RoToR-Freq.	53.1	46.4	53.7		—		32.3	29.2	36.1	33.5	29.5	37.2		—	
After shuffling segments, averaged over 3 seeds															
Original	49.7	43.5	51.0	62.8	58.5	64.5	30.3	27.6	35.0	31.6	27.9	35.5	42.1	38.9	44.7
↪ stdev. (±)	0.34 / 0.28 / 0.46			0.29 / 0.28 / 0.05			0.26 / 0.24 / 0.35			0.40 / 0.56 / 0.42			0.30 / 0.40 / 0.35		
PINE	51.8	45.3	52.7	64.3	59.8	65.9	31.5	28.7	35.3	31.7	28.2	35.7	48.0	44.3	50.0
↪ stdev. (±)	0.15 / 0.16 / 0.19			0.16 / 0.15 / 0.14			0.17 / 0.20 / 0.21			0.18 / 0.16 / 0.14			0.02 / 0.04 / 0.05		
RoToR	52.7	45.9	53.5	64.5	60.0	66.1	32.5	29.6	36.1	34.2	30.1	38.0	48.3	44.3	50.3
↪ stdev. (±)	0.05 / 0.09 / 0.04			0.02 / 0.02 / 0.01			0.11 / 0.06 / 0.09			0.06 / 0.05 / 0.04			0.05 / 0.09 / 0.05		
RoToR-MonoT5	52.2	45.2	52.8		—		32.3	29.4	35.9	32.8	28.8	36.5		—	
↪ stdev. (±)	0.16 / 0.18 / 0.18			—			0.16 / 0.13 / 0.07			0.16 / 0.09 / 0.07			—		
RoToR-Freq.	53.1	46.4	53.7		—		32.4	29.3	36.1	33.7	29.6	37.4		—	18
↪ stdev. (±)	0.02 / 0.07 / 0.03			—			0.09 / 0.04 / 0.06			0.04 / 0.16 / 0.22			—		

Results: MMLU (selective routing)

Method	Llama-3.1-8B-Instruct			Qwen1.5-4B-Chat			Qwen1.5-7B-Chat		
	Init.	Rev.	Avg.	Init.	Rev.	Avg.	Init.	Rev.	Avg.
Orig.	68.3	64.8	65.5 $\pm$ 1.0	53.6	51.9	52.6 $\pm$ 0.6	60.1	56.6	58.6 $\pm$ 0.9
PCW	57.0	55.1	56.1 $\pm$ 1.1		–			–	
Set-Based Prompting	31.1	33.0	31.6 $\pm$ 0.8		–			–	
PINE	64.8	63.3	63.6 $\pm$ 0.7	50.5	49.3	49.4 $\pm$ 0.5	57.0	54.4	55.8 $\pm$ 0.9
RoToR	63.2	62.6	62.8 $\pm$ 0.7	49.6	47.7	48.3 $\pm$ 0.7	56.5	55.8	56.2 $\pm$ 0.6
$\hookrightarrow$ + S.R.	68.5	65.1	65.7 $\pm$ 0.9	53.7	51.8	52.6 $\pm$ 0.6	60.1	57.4	58.8 $\pm$ 0.7
RoToR - MonoT5	64.2	62.9	63.5 $\pm$ 0.5	49.7	47.6	48.7 $\pm$ 0.7	56.2	54.4	55.5 $\pm$ 0.7
$\hookrightarrow$ + S.R.	68.4	65.2	65.8 $\pm$ 0.9	53.8	51.9	52.6 $\pm$ 0.6	60.1	57.3	58.7 $\pm$ 0.8
RoToR - Freq.	64.3	63.6	63.8 $\pm$ 0.6	49.9	47.6	48.7 $\pm$ 0.5	56.4	54.7	55.7 $\pm$ 0.7
$\hookrightarrow$ + S.R.	68.5	65.3	65.8 $\pm$ 0.8	53.7	52.3	52.6 $\pm$ 0.6	60.0	57.3	58.6 $\pm$ 0.8
RoToR + S.R. (Oracle)	75.0	71.9	72.7 $\pm$ 1.0	61.8	60.1	61.1 $\pm$ 1.0	68.1	66.2	67.2 $\pm$ 0.7

- Order-invariant models fail (than the original model) with single use (expected)
- Selective Routing shows improved performance and stability across input re-orderings
- High S.R. (Oracle) value indicates high potential for further accuracy gains by optimizing choices on routing methods

Summary

We propose RoToR: a simple, effective order-invariant LM that..

- Can be applied to **any** zero-shot decoder-only model (with RoPE)
- Global sort + circular IDs mitigate positional bias
- Selective Routing enables practical use
- Paper: <https://arxiv.org/pdf/2502.08662>
- Code: github.com/soyoung97/RoToR



Code



Paper

Thank you!

Appendix

Appendix: computational overhead

PINE requires two additional operations:

(1) computing attention scores without rotary position embeddings ($\mathcal{O}(n^2d)$) and (2) sorting k segments for each query token ($\mathcal{O}(nk \log k)$), totaling $\mathcal{O}(n^2d + nk \log k)$ (Wang et al., 2024)⁴.

our lexicographical sorting requires only a single global sort of k segments ($\mathcal{O}(k \log k)$), each with length $\mathcal{O}(n)$, achieving $\mathcal{O}(nk \log k)$ and eliminating the expensive $\mathcal{O}(n^2d)$ term entirely. This can be further optimized to $\mathcal{O}(nk)$ using radix sort.⁵

Appendix: Example Input/Output

lost in the middle

Prefix:

<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe. Please ensure that your responses are socially unbiased and positive in nature. If a question does not make any sense, or is not factually coherent, explain why instead of answering something not correct. If you don't know the answer to a question, please don't share false information.<|eot_id|><|start_header_id|>user<|end_header_id|>

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Parallel texts:

Document [1](Title: List of Nobel laureates in Physics) The first ...

...

Document [10](Title: Nobel Prize in Chemistry) on December 10, the ...

Suffix:

Question: who got the first nobel prize in physics<|eot_id|><|start_header_id|>assistant<|end_header_id|>

Figure 7: Example input for the lost in the middle dataset.

Appendix: Example Input/Output

Mintaka

Prefix:
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

Below are the facts in the form of the triple meaningful to answer the question. Answer the given question in a JSON format, such as "Answer": "xxx". Only output the JSON, do NOT say any word or explain.

<|eot_id|><|start_header_id|>user<|end_header_id|>

Parallel texts:
(Super Bowl XLII, winner, New York Giants)
(Super Bowl XLII, participating team, New York Giants)
(Super Bowl XLII, point in time, time: +2008-02-03)
(Super Bowl XLII, followed by, Super Bowl XLIII)
(Super Bowl XLII, location, State Farm Stadium)
...
(Super Bowl XLII, sport, American football)
(Super Bowl XLII, instance of, Super Bowl)

Suffix:
Question: which team did the super bowl xlii mvp play for?, Answer: <|eot_id|><|start_header_id|>
assistant <|end_header_id|>

Gold Answer(s):
(‘NYG’, ‘Giants’, ‘NY Giants’, ‘New York Giants’)

Example generated output:
{ "Answer": "New York Giants" } (Parsed to: New York Giants)

Figure 10: Example input for the Mintaka dataset.

Appendix: Example Input/Output

MMLU

Prefix:
The following are multiple choice questions (with answers) about moral disputes.

Norcross agrees that if a being is incapable of moral reasoning, at even the most basic level, then it cannot be

Parallel texts:
A. a being of value.
B. an object of moral sympathy.
C. a moral agent.
D. a moral patient.

Suffix:
Answer:

Figure 11: Example input for the MMLU benchmark.

Appendix: LongBench-2WikiMultiHopQA

Order	Method	Llama 3.1-8B-Instruct				Qwen 1.5-7B-Chat			
		0-4k	4-8k	8k+	Total	0-4k	4-8k	8k+	Total
	Count	25	131	144	300	23	121	156	300
Initial (e.g., 1,2,3,4,5)	Orig.	48.3	56.8	34.0	45.1	65.6	47.9	26.7	38.2
	PINE	51.0	47.6	–	–	70.2	45.1	–	–
	RoToR	59.0	52.7	41.8	48.0	75.7	47.8	31.0	41.2
Reversed (e.g., 5,4,3,2,1)	Orig.	57.0	51.5	39.0	46.0	53.4	43.3	34.2	39.3
	PINE	43.0	49.8	–	–	64.1	48.9	–	–
	RoToR	59.0	52.0	41.0	47.3	72.8	47.6	30.8	40.8
Center flip (e.g., 3,2,1,5,4)	Orig.	47.0	47.7	35.6	41.8	61.0	40.6	32.7	38.1
	PINE	46.3	49.2	–	–	70.2	43.5	–	–
	RoToR	59.0	52.5	41.5	47.8	77.1	47.3	30.9	41.0

Table 9: F1 scores (%) on LONGBENCH-2WikiMultihopQA with ~ 10 k-token contexts. “Count” is the number of examples per length bucket; “–” denotes out-of-memory.

Appendix: Selective Routing ratio

Sorting	Llama-3.1-8B-Instr.			Qwen1.5-4B-Chat			Qwen1.5-7B-Chat		
	Init.	Rev.	Avg.	Init.	Rev.	Avg.	Init.	Rev.	Avg.
Lexical	7.0	8.5	7.3 ± 0.8	5.9	6.2	6.2 ± 0.4	10.3	10.6	9.9 ± 0.6
MonoT5	6.9	7.6	6.7 ± 1.5	8.0	12.5	9.8 ± 2.1	10.7	10.9	10.7 ± 0.7
Freq.	6.4	6.7	6.9 ± 0.5	8.5	10.9	9.4 ± 1.6	10.7	11.1	11.1 ± 0.8

Table 11: Selection ratio (%) of the RoToR variant under SR. Higher values indicate more frequent routing to RoToR.